MATHEMATICAL ANALYSIS 1
HOMEWORK 7

(1) **Products of expansions.** Let $p_n(x) = a_0 + a_1 x + \cdots + a_n x^n$ and $q_m(x) = b_0 + b_1 x + \cdots + b_m x^m$ be polynomials of degrees n and m (so $a_n \neq 0$, $b_m \neq 0$).

(a) Write the product $p_n(x)\, q_m(x)$ in the form $\sum_k c_k x^k$: what is its degree, and what is the coefficient c_k? *(Pay attention: n and m might be different.) Hint: first, as an example, try expanding the product $(a_0+a_1x+a_2x^2)(b_0+b_1x)$ and collect the terms with x^0, x^1, x^2, x^3. Which products $a_i b_j$ contribute to the coefficient of x^k?*

(b) Suppose that, as $x \to 0$,

$$f(x) = p_n(x) + o(x^n) \qquad \text{and} \qquad g(x) = q_m(x) + o(x^m).$$

Let $N = \min\{n, m\}$. Show that $f(x)\, g(x) = r(x) + o(x^N)$ as $x \to 0$, where r is the polynomial obtained from $p_n\, q_m$ by deleting all the terms of degree larger than N. Name the rules (Proposition 8.1, Corollary 8.2) that you use at each step.

(c) If $f(x) = 1 + 2x - x^2 + o(x^2)$ and $g(x) = 3 - x + o(x)$ as $x \to 0$, write $f(x)\, g(x)$ in the same form. Can you determine the coefficient of x^2 in $f(x)\, g(x)$? Explain.

(2) **Limits with equivalences.** Compute the following limits, using the fundamental limits written as equivalences (Section 8.1.5) together with Proposition 8.3 and Corollary 8.4:

(a) $\displaystyle\lim_{x\to 0} \frac{\sin(3x)\, \ln(1+2x)}{x\,(e^x - 1)}$

(b) $\displaystyle\lim_{x\to 0} \frac{1-\cos(2x)}{x \ln(1+3x)}$

(c) $\displaystyle\lim_{x\to 0} \frac{\sqrt[3]{1+6x} - 1}{\sin(4x)}$

(d) $\displaystyle\lim_{x\to 0} \frac{e^{x^2} - 1}{1 - \cos x}$

(e) $\displaystyle\lim_{x\to 0} \frac{\ln(1+3x) + x^2}{2x + \sin^2 x}$

(f) $\displaystyle\lim_{x\to +\infty} x^2 \left(1 - \cos\frac{1}{x}\right)$

(g) $\displaystyle\lim_{x\to +\infty} x \ln\left(1 + \frac{2}{x}\right)$

(h) $\displaystyle\lim_{x\to +\infty} \left(1 + \frac{3}{x}\right)^{2x}$

(i) $\displaystyle\lim_{x\to \frac{\pi}{2}^-} \left(\frac{1}{\cos x} - \tan x\right)$

(j) $\displaystyle\lim_{n\to\infty} n\left(e^{1/n} - 1\right)$

(k) $\displaystyle\lim_{n\to\infty} n^2 \left(1 - \cos\frac{\pi}{n}\right)$

Hint for (i): take $y = \frac{\pi}{2} - x$.

(3) **Equivalences cannot be used in sums.** A student argues as follows: *since* $\sin(2x) \sim 2x$ *and* $\sin x \sim x$ *as* $x \to 0$,

$$\lim_{x\to 0} \frac{\sin(2x) - 2\sin x}{x^3} = \lim_{x\to 0} \frac{2x - 2x}{x^3} = 0.$$

(a) Explain what is wrong with this argument.

(b) Compute the limit correctly. *Hint:* $\sin(2x) = 2\sin x \cos x$.

(4) **The hierarchy of infinities.**

(a) Using the ordering of infinite functions and sequences (Section 8.2), compute:

(i) $\displaystyle\lim_{x\to+\infty} \frac{x^5 + e^x}{2e^x + x^2}$

(ii) $\displaystyle\lim_{x\to+\infty} \frac{x^3 + 2^x}{3^x + x^{10}}$

(iii) $\displaystyle\lim_{x\to+\infty} \frac{\ln x}{\sqrt{x}}$

(iv) $\displaystyle\lim_{x\to+\infty} \frac{\ln x}{x^2}$

(v) $\displaystyle\lim_{n\to\infty} \frac{n! + 3^n}{n^n + n^{10}}$

(b) *Running times of algorithms.* Four algorithms solve the same problem; on an input of size n they need 2^n, $n\log_2 n$, $n!$ and n^3 elementary operations, respectively. Order them from the fastest to the slowest for large n, and justify the order. A computer performs 10^9 operations per second: how long does each algorithm take for $n = 50$?

(5) **Order and principal part.**

(a) Determine the order and the principal part with respect to $\varphi(x) = \frac{1}{x}$ as $x \to +\infty$ of the function $f(x) = \sin\left(\sqrt{x^2-1} - x\right)$.

(b) As $x \to +\infty$, the function $f(x) = \ln\left(9 + \sin\frac{2}{x}\right) - 2\ln 3$ can be written as $f(x) = \frac{b}{x^\alpha} + o(x^{-\alpha})$. Find b and α.

(c) Determine the order and the principal part with respect to $\varphi(x) = x$ as $x \to 0$ of the function $f(x) = \frac{e^x}{1+x^2} - 1$.

(6) **Uniqueness of the order.** Let $f : \mathbb{R} \to \mathbb{R}$ be infinite or infinitesimal at $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$, and let φ be the corresponding comparison function of Section 8.3.

(a) State the definition of the *order* α of f at x_0 with respect to φ.

(b) Prove that the order is unique: if f has order α and also order β with respect to φ, then $\alpha = \beta$. *Hint: look at the quotient* $|\varphi|^\alpha / |\varphi|^\beta$.

(7) **More orders and principal parts.**

(a) Determine the order and the principal part of each of the following functions with respect to the indicated φ:

(i) $\sqrt{1+x^2} - 1$ as $x \to 0$, with respect to $\varphi(x) = x$;

(ii) $1 - \cos\sqrt{x}$ as $x \to 0^+$, with respect to $\varphi(x) = x$;

(iii) $x^3 + \sqrt{x}$ as $x \to 0^+$, and as $x \to +\infty$, with respect to $\varphi(x) = x$;

(iv) $\frac{x^2+1}{x-1}$ as $x \to 1$, with respect to $\varphi(x) = \frac{1}{x-1}$.

(b) *How far is the horizon?* From a height $h > 0$ above the sea, the distance to the horizon is $d(h) = \sqrt{(R+h)^2 - R^2}$, where $R \approx 6371$ km is the radius of the Earth (the line of sight is tangent to the Earth). Show that d is infinitesimal as $h \to 0^+$, and find its order and principal part with respect to $\varphi(h) = h$. Deduce the rule of thumb "$d \approx 3.57\sqrt{h}$ km, with h in meters" and use it for $h = 1.7$ m (standing on the beach) and $h = 100$ m (a lighthouse). How does $d(h)$ behave as $h \to +\infty$?

(8) **Asymptotes.**

(a) Find all the asymptotes (vertical, horizontal and oblique) of the following functions, on their natural domains:

(i) $\dfrac{x^2+3x-1}{x+2}$

(ii) $\sqrt{x^2+4x}$

(iii) $x\,e^{1/x}$

(iv) $x \arctan x$

(v) $e^{\sqrt{x}}$

(vi) $\dfrac{1}{(x^2+4)^2}$

(vii) $\ln(\sin x)$, $x \in (0, \pi)$

(viii) $\sqrt[3]{x^2-3x}$

(ix) $\arctan\dfrac{x}{x-1}$

Hint for (iv): for $x > 0$, $\frac{\pi}{2} - \arctan x = \arctan\frac{1}{x}$.

(b) *A falling body with air resistance.* A body of mass m falls from rest; the air exerts on it a resistance force proportional to its velocity, kv with $k > 0$. Physics shows that at time $t \geq 0$ its velocity and the distance it has covered are

$$v(t) = v_T\left(1 - e^{-t/\tau}\right), \qquad s(t) = v_T t - v_T\tau\left(1 - e^{-t/\tau}\right),$$

where $v_T = \frac{mg}{k}$ and $\tau = \frac{m}{k}$.

(i) Show that $\lim_{t\to+\infty} v(t) = v_T$. This is why v_T is called the *terminal velocity*: it is the speed at which the air resistance balances gravity.

(ii) Show that s has an oblique asymptote as $t \to +\infty$ and find it. Interpret the result: compared with a body that moves with constant velocity v_T from the start, how far behind is the falling body after a long time?

(9) **Zeros on an open interval.** In this problem we prove Corollary 9.2 of the lecture notes (the existence of a zero of a continuous function on an open interval (α, β), where α may be $-\infty$ and β may be $+\infty$). You are guided in the steps below.

(a) State Corollary 9.2.

(b) Suppose that $\ell_\alpha < 0 < \ell_\beta$ (state that the case $\ell_\alpha > 0 > \ell_\beta$ is similar, and is therefore omitted). Fix a point $c \in (\alpha, \beta)$.

(c) Using Theorem 6.2 (local sign, applied to the right limit at α), find a point $a \in (\alpha, c)$ with $f(a) < 0$. Treat separately the cases $\alpha \in \mathbb{R}$ and $\alpha = -\infty$.
(d) In the same way, find a point $b \in (c, \beta)$ with $f(b) > 0$.
(e) Explain why Theorem 9.1 (Bolzano's Theorem) can be applied to f on $[a, b]$, and conclude that f has a zero in (α, β).
(f) Prove that the zero is unique if f is strictly monotone on (α, β).

(10) **Bolzano's Theorem at work.**
(a) Show that the equation $x^5 + x - 1 = 0$ has exactly one real solution, and that it lies in $(0, 1)$.
(b) Show that the equation $\cos x = x$ has exactly one solution in $[0, \frac{\pi}{2}]$.
(c) Show that every polynomial of odd degree has at least one real zero.
(d) *Bisection.* Perform three steps of the bisection method (as in the proof of Theorem 9.1) for $f(x) = x^3 - 2$ on $[1, 2]$, and give an interval of length $\frac{1}{8}$ containing $\sqrt[3]{2}$. How many steps guarantee that the midpoint of the last interval approximates $\sqrt[3]{2}$ with an error smaller than 10^{-3}?
(e) The function $f(x) = \frac{1}{x}$ satisfies $f(-1) < 0 < f(1)$, but it has no zero in $[-1, 1]$. Which hypothesis of Bolzano's Theorem fails?
(f) *Kepler's equation.* The position of a planet or a satellite on its elliptic orbit at time t is found by solving Kepler's equation

$$M = E - e \sin E$$

for the unknown $E \in \mathbb{R}$, where $M \in \mathbb{R}$ is proportional to t and $e \in [0, 1)$ is the eccentricity of the orbit. Show that for every $M \in \mathbb{R}$ the equation has at least one solution $E \in \mathbb{R}$. *(It is in fact unique, but we will prove this later.)*

(11) **Proofs with the Intermediate Value Theorem and Weierstrass' Theorem.**
(a) The lecture notes prove the Intermediate Value Theorem (Theorem 9.4) only in the case $f(a) < f(b)$. Prove the case $f(a) > f(b)$.
(b) Let $f : [0, 1] \to [0, 1]$ be continuous. Prove that f has a *fixed point*: there exists $x_0 \in [0, 1]$ such that $f(x_0) = x_0$. *Hint: consider* $g(x) = f(x) - x$.
(c) Let f be continuous on $[a, b]$ with $f(x) > 0$ for all $x \in [a, b]$. Prove that there exists $m > 0$ such that $f(x) \geq m$ for all $x \in [a, b]$. Is the same true on $(0, 1]$ instead of $[a, b]$?
(d) Let $f : \mathbb{R} \to \mathbb{R}$ be continuous with $\lim_{x \to \pm\infty} f(x) = 0$ and $f(0) > 0$. Prove that f attains a maximum on $\mathbb{R}$.
(e) Let f be continuous on an interval I and suppose that $f(x) \in \mathbb{Q}$ for all $x \in I$. Prove that f is constant on I. *Hint: if $r < s$ are rational, then $r + \frac{s-r}{\sqrt{2}}$ is an irrational number between r and s.*

(12) **The Intermediate Value Theorem in practice.**
(a) A hiker starts walking up a mountain path at 8:00 and reaches the summit at 17:00. The next day she starts walking down the same path at 8:00 and reaches the bottom at 17:00 (her speed may vary, and she may stop to rest). Prove that there is a point of the path that she passes at the same time of the day on both days. *Hint: Corollary 9.3.*
(b) The temperature along a circular ring (for example, along the equator) is a continuous function $T(\theta)$ of the angle θ, with $T(\theta + 2\pi) = T(\theta)$. Prove that there are two diametrically opposite points of the ring with the same temperature. *Hint: consider* $g(\theta) = T(\theta) - T(\theta + \pi)$ *on* $[0, \pi]$.